

MASS-APP-014-V09 - Automation, Workflow & Intelligent Execution

Document Control

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1. Purpose

V09 defines how an approved human decision becomes a governed execution plan, coordinated work requests, monitored outcomes, and preserved learning.

V09 plans and supervises application-level execution. It does not execute workflows, dispatch personnel, authorize spending, publish communications, or replace operational owners.

2. Platform Boundaries

Owner	Responsibility
APP-014-V09	Decision-to-execution translation, execution plan, dependencies, approval gates, handoffs, monitoring view, exception intelligence
ENG-006	Durable workflow and task execution
ENG-020	Enterprise operations coordination
ENG-021	Mission and field dispatch
ENG-022	Financial transactions and expenditure authority
ENG-023/010	Communication governance and delivery
ENG-025	Automation governance and approval
ENG-009	AI and agent advisory execution

Published events: execution.plan.created, execution.plan.approved, execution.handoff.created, execution.exception.detected, execution.outcome.verified. Consumed events: decision.human.recorded, workflow.state.changed, mission.state.changed, automation.exception.raised, financial.authorization.updated.

3. Role Mapping

V09 roles extend and specialize the APP-014 hierarchy.

V09 role	Baseline	Specialized authority
Execution Owner	Steward, extending V01 Editor	Owns assigned plans, dependencies, handoffs, monitoring, and exception responses. Cannot approve its own plan when separation is required.
Execution Approver	Administrator, mapping to V01 Admin	Approves defined execution plans and amendments after ENG-004 confirms scope. It does not grant unrestricted administration.

Existing roles remain unchanged. Assignment narrows authority.

4. Lifecycle

Draft -> Validating -> Awaiting Approval -> Approved -> Handed Off -> Running -> Exception or Monitoring -> Outcome Verification -> Completed -> Archived.

The source HumanDecision is immutable. Plan amendments never rewrite the source decision. Material scope change returns to V03 for a new or amended human decision.

5. Capabilities

- Decision translation: derive objectives, constraints, conditions, owners, deadlines, and prohibited actions.
- Execution planning: define work packages, dependencies, required capabilities, resources, and success criteria.
- Workflow handoff: submit approved definitions to ENG-006.
- Operations handoff: submit coordinated work context to ENG-020.
- Dispatch handoff: submit mission requests to ENG-021 without assigning personnel.
- Automation handoff: submit approved automation requests to ENG-025.
- Financial gate: require ENG-022 authorization reference before cost-bearing work.
- Communication gate: route communication requirements through ENG-023.
- Monitoring: consume execution status without becoming system of record.
- Exception intelligence: explain blockers, impact, options, and required owner.
- Adaptive proposal: recommend a plan amendment; never apply it autonomously.
- Outcome verification: compare expected and verified results and preserve learning through V02.

V1 uses explicit plans, deterministic dependencies, typed handoffs, human approvals, status aggregation, and exception proposals. Future evolution may add policy-bounded dynamic replanning while retaining human authority.

6. Execution Plan

ExecutionPlan contains the decision reference, objective, conditions, success criteria, owner, time horizon, risk level, and required approvals.

WorkPackage contains one accountable owner, capability requirement, input contract, output contract, deadline, and completion criteria. Dependency edges are acyclic and tenant-local.

No package starts until prerequisite gates and dependencies are satisfied.

7. Approval and Adaptation

ExecutionApproval is immutable and records approver, authorization, decision, rationale, and conditions. Self-approval is prohibited where the requester and approver are the same.

ExecutionAmendmentProposal records trigger, current plan hash, proposed change, impact, risk, cost, evidence, and required authority. Approved amendments create a new plan revision.

Agents, NOVA, and POPS may advise through ENG-009. They cannot approve or invoke operational execution.

8. Monitoring and Exceptions

ExecutionStatusReference stores owning-system identifiers, status, version, observed time, and freshness. It is not the authoritative workflow or mission state.

ExecutionException records source, category, severity, affected packages, impact, evidence, owner, and resolution status. V09 presents practical options and routes the exception to its owner.

9. Security

- Every entity is tenant-bound and protected by RLS.
- Composite foreign keys protect child records.
- Decisions, approvals, status observations, and verified outcomes are immutable.

- Handoffs are idempotent and destination-scoped.
- Financial, communication, dispatch, and automation authority references are mandatory where applicable.
- Cross-tenant execution and learning are prohibited.

10. Data Model

Entity	Purpose
ExecutionPlan	Stable execution container
ExecutionPlanRevision	Immutable plan revision
WorkPackage	Bounded accountable work
WorkDependency	Acyclic package dependency
ExecutionGate	Required authorization or condition
ExecutionApproval	Immutable human approval
ExecutionHandoff	Idempotent owning-system request
ExecutionStatusReference	Read-only status observation
ExecutionException	Governed exception record
AmendmentProposal	Proposed plan revision
ResourceRequirement	Required capacity or resource
ExecutionOutcome	Human-verified result
ExecutionAuditEvent	Append-only lifecycle evidence

11. API Inventory

Method	Route	Purpose
GET, POST	/execution-plans	List or create plans
GET, PATCH	/execution-plans/{id}	Retrieve or edit draft plan
GET, POST	/execution-plans/{id}/revisions	List or create revisions
GET, POST	/execution-plans/{id}/work-packages	List or create packages
POST	/execution-plans/{id}/dependencies	Create dependency
DELETE	/execution-plans/{id}/dependencies/{dependencyId}	Remove draft dependency
GET, POST	/execution-plans/{id}/gates	List or create gates
POST	/execution-plans/{id}/approve	Record approval
POST	/execution-plans/{id}/handoffs/workflow	Handoff to ENG-006
POST	/execution-plans/{id}/handoffs/operations	Handoff to ENG-020
POST	/execution-plans/{id}/handoffs/dispatch	Handoff to ENG-021
POST	/execution-plans/{id}/handoffs/automation	Handoff to ENG-025
GET	/execution-plans/{id}/status	Aggregate status
GET	/execution-plans/{id}/exceptions	List exceptions
POST	/execution-plans/{id}/exceptions/{exceptionId}/options	Request advisory options
GET, POST	/execution-plans/{id}/amendments	List or propose amendments
POST	/amendments/{id}/approve	Approve amendment
POST	/execution-plans/{id}/outcomes	Record verified outcome
POST	/execution-plans/{id}/complete	Complete plan
POST	/execution-plans/{id}/archive	Archive plan

All mutations require idempotency keys.

12. Migration Reference

MASS-APP-014-V09_Migration_Reference.sql defines tables, columns, foreign keys, indexes, constraints, triggers, and RLS policies.

13. Failure Behavior

Invalid decision authority blocks plan creation. Missing gate blocks handoff. Dependency cycles are rejected. Downstream failures retain retry-safe handoffs. Stale status is displayed as unverified. Material amendments return to human approval.

14. Constitutional Boundary Statement

V09 coordinates approved intent into execution contracts. It does not own workflows, operations, dispatch, finance, communications, automation, or executive authority. Owning systems execute; accountable humans approve.